

# SooHyun Yang

Google Scholar: <https://scholar.google.com/citations?user=zczJPkAAAAJ&hl=en>

Github: <https://github.com/Yang-SooHyun>

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## EDUCATION

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### University of Seoul

*Master of Science in Environmental Engineering*

GPA: 4.4/4.5

Thesis: Spatiotemporal Responses of Global Lake Algal Blooms to Drought: An Analysis Using a High-Resolution Chlorophyll-*a* Dataset (Tentative)

Advisor: YoonKyung Cha

Seoul, South Korea

Sep. 2025 – Feb. 2027

### University of Seoul

*Bachelor of Science in Environmental Engineering*

GPA: 3.8/4.5

Thesis: Development of Bayesian Network for Probabilistic Mid-to Long-term Forecasting for Harmful Algal Blooms

Seoul, South Korea

Mar. 2020 – Aug. 2025

## RESEARCH INTERESTS

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Processing satellite observations, implementing robust quality control frameworks, and constructing long-term high-resolution geospatial datasets.

Analyzing long-term variability, identifying trends on the Earth's surface, and modeling spatiotemporal environmental dynamics under climate conditions.

Applying advanced neural network architectures for spatiotemporal data imputation, time-series reconstruction, and predictive environmental modeling.

Leveraging spatial deep learning, semantic segmentation, and object detection for automated feature extraction, land classification, and surface anomaly detection.

## RESEARCH EXPERIENCE

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### University of Seoul

*Graduate Research Assistant, Water Environment Management Lab*

Seoul, South Korea

Sep. 2025 – Feb. 2027

- Developed a cyanobacteria cell counts forecasting model utilizing deep learning models (Transformer, TFT, TCN, LCA).
- Imputed daily water quality and cyanobacteria cell counts data from mixed-frequency inputs using SAITS (Self-Attention-based Imputation for Time Series).
- Developed a transformer-based hierarchical deep learning model for satellite Chlorophyll-*a* prediction.
- Calibrated and adapted established bio-optical algorithms to optimize Chlorophyll-*a* retrieval accuracy for the unique optical properties of diverse global lakes.
- Constructed GLCD250-MOD, a daily, 250 m Chlorophyll-*a* dataset covering 465,966 inland lakes worldwide from 2000–2024.
- Developed a large-scale pipeline to acquire and process hundreds of terabytes of Terra MODIS imagery, including parallel downloading and polygon-based ROI extraction.
- Analyzed global spatiotemporal trends using GLCD250-MOD and evaluated the response and interaction mechanisms of lake ecosystems with conventional environmental variables under climate change.

*Undergraduate Research Assistant, Water Environment Management Lab*

Sep. 2023 – Aug. 2025

- Conducted a long-term spatiotemporal trend analysis using the Mann-Kendall test.
- Constructed a Lake Erie Chlorophyll-*a* dataset using open-source data from in-situ and satellites (ENVISAT MERIS, Aqua/Terra MODIS, Sentinel-3A/B OLCI, S-NPP/NOAA-20 VIIRS).
- Projected mid-to long-term algal blooms using Shared Socioeconomic Pathways (SSP).
- Developed a prediction model for cyanobacteria cell counts using Bayesian networks.
- Created a web-specific automatic crawling code.

## TEACHING AND MENTORING EXPERIENCE

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### Engineering Statistics

Teaching Assistant

Mar. 2026 – Jun. 2026

Answered course-related questions, created examples and homework programming codes.

### Graduation Thesis Class for Undergraduate Students

Academic Mentor

2026

Guided three undergraduates to successfully complete their bachelor's thesis.

- Thesis 2026: Correlation Analysis between Drought Indices and Algal Bloom Indices for Global Lakes

## HONORS & AWARDS

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### Graduate

Outstanding Oral Presentation Award

Feb. 2026

Graduate Student Research Competency Enhancement Program, Spring

Apr. 2026 – Jul. 2026

Instruction Assistance Scholarship

Mar. 2026 – Jun. 2026

The Combined BS/MS Degree Scholarship (Full tuition)

Sep. 2025 – Feb. 2027

### Undergraduate

Outstanding Conference Poster Award

Feb. 2025

Outstanding Graduate Thesis Award

Feb. 2025

Grand Prize, Innovative Talent Cultivation Project for Digital Water Industry

Dec. 2024

Student Future Planning Semester: Research Internship

Jul. 2024 – Aug. 2024

Innovative Talent Cultivation Project for Digital Water Industry

Sep. 2023 – Dec. 2024

Student Future Planning Semester: Global Self-Directed Research

Sep. 2023 – Dec. 2023

## PUBLICATIONS

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- [1] **Yang, S.**<sup>1</sup>, Lee, H.<sup>1</sup>, Lee, G., Gu, T., Park, T., Park, J., Shin, J., Kim, T., Cha, Y. (2026). GLCD250-MOD: Global Daily 250 m Chlorophyll-*a* Dataset for Inland Lakes based on MODIS, *Earth System Science Data (In preparation)*.
- [2] Park, T., Jung, J., Park, J., **Yang, S.**, Moon, J., & Cha, Y. (2026). Structurally Constrained Hierarchical Learning for Improving River Health Assessment using Tabular Transformers. *Water Research*, 126872.
- [3] Kim, T., Lee, H., **Yang, S.**, Lee, G., Shin, J., & Cha, Y. (2025). Long-term Spatiotemporal Variability and Regime Classification of Chlorophyll-*a* Concentrations in Lake Erie Using Satellite Products. *Harmful Algae*, 102896.

## CONFERENCES

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- [1] **Yang, S.**, Lee, H., Kim, T., & Cha, Y. Long-term Trend Analysis of Algal Blooms in Global Lakes. International Water Association World Water Congress & Exhibition, 2026.
- [2] Lee, H., **Yang, S.**, Kim, T., & Cha, Y. Construction of a Long-term High Spatiotemporal Resolution Chlorophyll-*a* Dataset for Global Lakes. Korean Society on Water Environment – Korean Society of Water & Wastewater Joint Spring Conference, 2026.
- [3] **Yang, S.**, Lee, H., Park, T., Gu, T., Moon, J., Park, S., & Cha, Y. Advancement of Aquatic Ecosystem Model Development and Integration Strategies within a Digital Twin Korea-Water. Korean Society of Limnology Winter Conference, 2026.
- [4] Kim, T., Lee, H., **Yang, S.**, & Cha, Y. Does drought intensify algal blooms in global lakes? American Geophysical Union Winter Conference, 2025.
- [5] Lee, H., **Yang, S.**, Jeong, J., Hwang, G., Moon, J., & Cha, Y. Development of a Machine Learning-Based Geographically Weighted Model for Predicting a Integrated Assessment Index of River Health. Korean Society on Water Environment – Korean Society of Water & Wastewater Joint Fall Conference, 2025.
- [6] **Yang, S.**, Park, T., Park, J., Gu, T., & Cha, Y. Habitat Suitability Assessment and Derivation of Optimal Environmental Conditions in the Geum and Nakdong River Basins Using a Data-Driven HSI Model. Korean Society of Limnology Summer Conference, 2025.
- [7] **Yang, S.**, Gu, T., Park, T., Lee, H., & Cha, Y. Development of Mid-to Long-term Forecasting Model of

Harmful Algal Blooms Based on Probabilistic Predictions Using a Bayesian Network. Korean Society of Limnology Winter Conference, 2025.

- [8] Kim, T., Lee, G., Lee, H., **Yang, S.**, Shin, J., & Cha, Y. Characterizing Long-term Spatiotemporal Variability and Regime Heterogeneity of Chlorophyll-a in Lake Erie Using Multiple Satellite-based Products. American Geophysical Union Winter Conference, 2024.
- [9] **Yang, S.**, Lee, H., Kim, T., Shin, J., & Cha, Y. Spatiotemporal long-term pattern analysis and regime classification in Lake Erie using an optimal satellite product. Korean Society of Environmental Engineers Winter Conference, 2024.
- [10] **Yang, S.**, Lee, H., Lee, G., & Cha, Y. Application of Explainable Deep Learning Model for River Eutrophication Assessment. Applied Artificial Intelligence Conference, 2024.
- [11] **Yang, S.**, Jeong, B., & Cha, Y. Analysis of the relationship between the exogenous factor to the Harmful Algae Blooms in the Sejong Weir based on PLS-SEM. Applied Artificial Intelligence Conference, 2023.

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## PROJECTS

- [1] Green Resource Engineering and Advanced Transformation (GREAT) Center for Waste-to-Wealth, National Research Foundation, Graduate Researcher, Jul. 2026 –Present
- [2] Development of a Mechanism-Based River Water Quality Model Integrated with Artificial Intelligence (II), National Institute of Environmental Research (NIER), Graduate Researcher, Jun. 2026 – Mar. 2027.
- [3] Digital Twin Korea-Water Development Project (5/5): A Study on Optimal Habitat Environmental Conditions for the Application of Ecological Models (II), National Institute of Environmental Research (NIER), Graduate Researcher, Mar. 2026 – Jan. 2027.
- [4] Development of a Mechanism-Based River Water Quality Model Integrated with Artificial Intelligence (I), National Institute of Environmental Research (NIER), Graduate Researcher, May 2025 – Feb. 2026.
- [5] Digital Twin Korea-Water Development Project (4/5): A Study on Optimal Habitat Environmental Conditions for the Application of Ecological Models (I), National Institute of Environmental Research (NIER), Graduate Researcher, Mar. 2025 – Dec. 2025.
- [6] Development and verification of digital twin-based water quality prediction, National Information Society Agency (NIA), Undergraduate Research Assistant, May. 2024 – Dec. 2024.
- [7] Development and Pilot Application of a Digital Twin Aquatic Ecosystem Model (II), National Institute of Environmental Research (NIER), Undergraduate Research Assistant, Mar. 2024 – Dec. 2024.
- [8] Development of a modularized AI-based model for the multiscale analysis of multimodal environmental data, Ministry of Science and ICT (MSIT), Undergraduate Research Assistant, Mar. 2024 – Feb. 2025.
- [9] Water Policy Outlook Based on Integrated Analysis of Water, Environment, and Economy(I), Korea Environment Institute (KEI), Undergraduate Research Assistant, Mar. 2024 – Dec. 2024.

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## SKILLS

### Software

Python, R, ArcGIS, Google Earth Engine, Netica, & Crawling.

### Modeling Techniques

Bayesian statistics, Bayesian network, Structural equation modeling, Deep learning, & Machine learning.

### Languages

Korean (native), English (advanced), Spanish (elementary), & Mandarin (elementary).

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## ACTIVITIES & INTERESTS

### Falcons, University of Seoul baseball club

*Club Member*

Sep. 2020 – Present

Played baseball as a second baseman and pitcher.

### Other Interests

Baseball, Flute, and Trekking

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## MILITARY SERVICE

Republic of Korea Army

Jan. 2021 – Jul. 2022

Completed military service as a drill instructor for basic military training.

## REFERENCES

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YoonKyung Cha, Professor  
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